Upskilling Collaboration Project Catalog

Compact project menu for choosing a GitHub collaboration or learning project.

Pocket Cocktail Party

Repository: https://github.com/bballdave025/pocket-cocktail-party

Status: Prototype / active experiment

What It Is: Blind-source-separation project that starts with understandable synthetic mixtures and Independent Component Analysis, then moves toward messier real recordings. It keeps the mixing geometry visible instead of treating audio separation as a black box.

Tools and Concepts Used: Python; ICA/BSS, linear algebra, signal processing, synthetic data, experimental design.

Bilingual Boundary Lab

Repository: https://github.com/bballdave025/bilingual-boundary-lab

Status: Prototype / active experiment

What It Is: Uses only *War and Peace* to test how models from n-grams upward learn boundaries between Russian and embedded French. Extensions can compare explicit boundary tokens with weaker or implicit language-switch signals.

Tools and Concepts Used: Python; n-grams, multilingual NLP, tokenization, sequence modeling, visualization.

English Gorf

Repository: https://github.com/bballdave025/english-gorf

Status: Prototype with an inconclusive first experiment

What It Is: Tests whether English's Germanic and French history leaves detectable differences in letter-frequency distributions, initially comparing shorter and longer words. A collaborator could vary vocabulary size, similarity measures, visualization, or source-separation approaches.

Tools and Concepts Used: Python; frequency vectors, cosine similarity, dictionaries/corpora, visualization, computational linguistics.

Envelope ZIP Reading

Repository: https://github.com/bballdave025/envelope-zip-reading

Status: Scaffold / collaboration-ready

What It Is: MNIST digit-recognition project inspired by Michael Nielsen's online neural-network text. A good small collaboration would implement the simple hidden-layer model and CNN, then inspect individual mistakes rather than stopping at aggregate accuracy.

Tools and Concepts Used: Python; neural networks, CNNs, MNIST, classification, error analysis, linear algebra.

JabberVec-l2v

Repository: https://github.com/bballdave025/jabbervec-l2v

Status: Prototype / active experiment

What It Is: Character-level modeling experiment asking what geometry learned letter embeddings develop from English-like text. It is intentionally small enough to inspect embeddings, generation behavior, and low-rank structure directly.

Tools and Concepts Used: Python/PyTorch; embeddings, PCA/SVD, character models, low-rank approximation, visualization.

Latent Forge

Repository: https://github.com/bballdave025/latent-forge

Status: Active experiment

What It Is: Small recommender-system laboratory built around simple baselines, matrix factorization, and holdout MSE. The emphasis is on changing one assumption at a time and justifying added complexity.

Tools and Concepts Used: Python; recommender systems, matrix factorization, low-rank models, linear algebra, evaluation.

Nyquist Text Existence Criterion

Repository: https://github.com/bballdave025/nyquist-text-existence

Status: Active preregistered research project

What It Is: Investigates when writing ceases to exist in a sampled image as resolution decreases, distinct from whether OCR can still recognize it. The project is built around measurable structural collapse under downsampling.

Tools and Concepts Used: Python; sampling theory, image processing, information theory, preregistration, experimental design.

Manuscript Waste Reuse Finder / RMFB

Repository: https://github.com/bballdave025/manuscript-waste-reuse-finder

Status: Published early phase / active research lineage

What It Is: Uses computer vision to find reused manuscript fragments in historical book bindings. The project combines unusual dataset construction, archival research, classification, provenance, and manuscript studies.

Tools and Concepts Used: Python; computer vision, transfer learning, ResNet50, dataset curation, provenance, historical documents.

RIB-WRIST-IN-BIN-DIN / RMMFB Planning

Repository: https://github.com/bballdave025/rib-wrist-in-bin-din

Status: Active research/planning

What It Is: Later RMMFB research home for model planning, paper ideas, high-resolution image work, explainability, and experiment design.

Tools and Concepts Used: Python/Jupyter; CV, high-resolution models, Grad-CAM, transfer learning, research design.

FHTW Paper Code Prep

Repository: https://github.com/bballdave025/fhtw-paper-code-prep

Status: Working experiment infrastructure

What It Is: Cross-platform experiment scaffold developed while preparing RMMFB model work, with notebook creation, reproducibility/logging, CIFAR-10 proving experiments, and AWS/SageMaker preparation.

Tools and Concepts Used: Bash, PowerShell, Python/Jupyter; experiment scaffolding, testing, reproducibility, AWS/SageMaker.

Binding Unwinding

Repository: https://github.com/bballdave025/binding-unwinding

Status: Working data-preparation tool

What It Is: Extracts and normalizes manuscript/document images from PDFs for later ML use. It is a practical example of the data-engineering work that often dominates real computer-vision projects.

Tools and Concepts Used: Python, Bash, PyMuPDF, ImageMagick; batch processing, normalization, reproducibility.

RMMFB Ontology Documentation

Repository: https://github.com/bballdave025/congenial-chainsaw-rmfb-html

Status: Active documentation / historical prototype

What It Is: Began as an overbuilt HTML classification helper, then evolved into the main home for concise Markdown/PDF ontology documentation and classification-change history.

Tools and Concepts Used: Markdown/HTML; ontology design, annotation systems, documentation, versioned classification rules.

Text Cleanroom

Repository: https://github.com/bballdave025/text-cleanroom

Status: Active tooling project

What It Is: Deterministic toolkit for finding and repairing messy text and filename encodings while keeping changes auditable.

Tools and Concepts Used: Python; Unicode/encoding, CLI design, preprocessing, data quality, testing.

Multimodal Jupy Logger

Repository: https://github.com/bballdave025/multimodal-jupy-logger

Status: Experimental tooling

What It Is: Captures Jupyter/IPython work as durable multimodal logs that can later be replayed or rendered, emphasizing experiment provenance rather than only notebook state.

Tools and Concepts Used: Python/Jupyter; MIME handling, event logging, provenance, reproducibility.

No-Code Jupyter Notebook Tools

Repository: https://github.com/bballdave025/no-code-jupyter-nb

Status: Working toolkit

What It Is: Adds dropdowns, file selectors, dialogs, clipboard helpers, and other low-code controls inside Jupyter for non-programmer workflows.

Tools and Concepts Used: Python/Jupyter; UI widgets, OOP, package design, human-centered tooling.

Mendeleev Spelling Bee

Repository: https://github.com/bballdave025/mendeleev-spelling-bee

Status: Working prototype

What It Is: Multilingual word game that tests whether words can be represented as sequences of chemical-element symbols, extending beyond English into Cyrillic and other scripts.

Tools and Concepts Used: Python; Unicode, tokenization, search/segmentation, CLI design, pytest/TDD.

AWS Notebook Typings

Repository: https://github.com/bballdave025/dwb-aws-nb-typings

Status: Learning/project-mining repository

What It Is: Re-created selected AWS Machine Learning Specialty course notebooks by typing and understanding the code rather than simply cloning it. The collaboration use is extracting fresh mini-project ideas from the concepts without redistributing course code.

Tools and Concepts Used: Python, NumPy, AWS; ML workflows, data wrangling, notebook-based learning, cloud ML.

NLP with PyTorch Project Mining

Repository: https://github.com/bballdave025/nlp_w_pytorch_zhongyu-pan

Status: Coursework-derived project seed

What It Is: Includes a CNN sentiment-analysis project from an NLP/PyTorch course. A collaborator could identify the learning objective and build a new, independent text-classification project around the same concepts.

Tools and Concepts Used: Python/PyTorch; CNNs, sentiment analysis, NLP, supervised learning, text preprocessing.

Start Pirate Now

Repository: https://github.com/bballdave025/start-pirate-now

Status: Small learning experiment

What It Is: Started as interview preparation for R by systematically working through the language's basic command-line/help documentation. The same idea generalizes to learning another compact tool or language from its canonical documentation.

Tools and Concepts Used: R; documentation-driven learning, scripting, basic data structures.

Solid Rotary Phone

Repository: https://github.com/bballdave025/solid-rotary-phone

Status: Coursework-derived project seed

What It Is: Material from a strong TensorFlow Developer Certificate course containing several useful project patterns. The collaboration opportunity is to extract problem statements and rebuild fresh versions rather than reuse course solutions.

Tools and Concepts Used: Python/TensorFlow; neural networks, CNNs, training/evaluation, ML workflow.

Upgraded Waffle

Repository: https://github.com/bballdave025/upgraded-waffle

Status: Coursework-derived project seed

What It Is: Broad TensorFlow course work covering model families through Transformers, including an unusually detailed exploration of TensorFlow computation graphs.

Tools and Concepts Used: Python/TensorFlow; computation graphs, neural networks, sequence models, Transformers.

Double-P Double-N

Repository: https://github.com/bballdave025/double-p-double-n

Status: Minimal prototype / larger research seed

What It Is: First small implementation of the Paper/Pencil Neural Networks idea: physically inspectable neural networks that can scale from pedagogy toward questions about computational and representational cost.

Tools and Concepts Used: Neural networks, linear algebra, computational complexity, information theory, hands-on visualization.

Discrete Fourier Transforms and Sampling

Repository: https://github.com/bballdave025/discrete_fourier_transforms_and_sampling

Status: Small teaching/scientific-computing project

What It Is: Short Mathematica-based introduction to Fourier analysis and sampling originally used to explain concepts relevant to ASR work. It could be recreated in Python or expanded into a signal-processing project.

Tools and Concepts Used: Wolfram Language/Mathematica; DFT/FFT, sampling, signal processing, computational mathematics.

Higgs Boson Visualized

Repository: https://github.com/bballdave025/higgs_boson_visualized

Status: Scientific-visualization project

What It Is: Particle-physics visualization work centered on Higgs-boson concepts, suitable as a starting point for reproducible quantitative physics communication.

Tools and Concepts Used: Scientific computing/visualization; particle physics, quantitative plotting.

RWKV + LoRA Learning Experiment

Repositories: https://github.com/bballdave025/rwkv-lora and https://github.com/bballdave025/info-rwkv-lora-learning-ref

Status: Experimental / ambiguous result

What It Is: Attempted LoRA fine-tuning work on RWKV, with supporting learning/reference material. The experiment produced behavior worth reconstructing rather than being presented as a clean success.

Tools and Concepts Used: Python; RWKV, LoRA/PEFT, sequence models, loss curves, training diagnostics.

Job Application Word Frequency

Repository: https://github.com/bballdave025/job-app-word-freq

Status: Small text-analysis tool

What It Is: Compares language in job descriptions and application materials using lightweight frequency analysis.

Tools and Concepts Used: Python; tokenization, word frequencies, text comparison, visualization.

DWB Python Utilities

Repository: https://github.com/bballdave025/dwb_python_util

Status: Utility library

What It Is: Reusable Python helpers accumulated across projects; suitable for refactoring, packaging, documentation, and testing practice.

Tools and Concepts Used: Python; package design, refactoring, testing, reusable components.

DWB Utilities

Repository: https://github.com/bballdave025/dwb_util

Status: Utility repository

What It Is: General-purpose support utilities accumulated across technical work, best suited to a small software-maintenance or cleanup collaboration.

Tools and Concepts Used: Scripting/software utilities; modularity, automation, refactoring, documentation.

uGTgtdD System

Repository: https://github.com/bballdave025/gtgtdd-sys

Status: Active personal-system tooling

What It Is: Personal GTD/task-management system with explicit workflow and context-document practices.

Tools and Concepts Used: Git/Markdown/scripting; workflow design, information retrieval, automation.

AWS Udemy Pacing

Repository: https://github.com/bballdave025/aws-udemy-pacing

Status: Working automation

What It Is: Generates pacing/progress plans for technical courses and certifications.

Tools and Concepts Used: Python; spreadsheet generation, date arithmetic, automation, progress tracking.

Didactic Spoon

Repository: https://github.com/bballdave025/didactic-spoon

Status: Coursework / project seed

What It Is: Deep-learning/Keras/TensorFlow study notebooks that can be mined for small independent experiments.

Tools and Concepts Used: Python, Keras/TensorFlow; neural networks, training loops, model evaluation.

Palmer Penguins

Repository: https://github.com/bballdave025/palmerpenguins

Status: Data-science demo / project seed

What It Is: Work around the Palmer Penguins dataset, useful as a baseline exercise before replacing the familiar dataset with something more original.

Tools and Concepts Used: Python/R; tabular data, visualization, classification/regression, exploratory analysis.

Visualization Portfolio

Repository: https://github.com/bballdave025/visualization_portfolio

Status: Portfolio collection

What It Is: Collection of data-visualization work that can be improved exhibit-by-exhibit or turned into polished explanatory graphics.

Tools and Concepts Used: Data visualization; plotting, exploratory analysis, scientific communication.

STEM Showcase

Repository: https://github.com/bballdave025/stem-showcase

Status: Demonstration collection

What It Is: General STEM showcase material from which individual pieces can be selected for focused science-communication or coding collaborations.

Tools and Concepts Used: STEM communication; visualization, programming, explanatory design.

Cyrillic-through-Latin OCR

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea / dataset-building experiment

What It Is: Build paired data by copy/pasting true Cyrillic text while separately using Android/Pixel screen-selection OCR to capture its Latin-biased corruption. The resulting pairs could train a model to reverse similar historical OCR errors.

Tools and Concepts Used: Mobile data collection; OCR, Unicode/Cyrillic, sequence transduction, noisy-channel modeling.

Multilingual Name Corruption Lab

Repository: Not yet created

Status: Idea / larger family-history experiment

What It Is: Simulate historical forms and mistakes for one surname, initially `KIMONT`, across languages, declensions, clerk transcription, typing, OCR, and eventually ASR. The goal is to generate better search variants for transformed historical names.

Tools and Concepts Used: Python/LLMs; synthetic data, transliteration, morphology, OCR/ASR error models, multilingual NLP.

DIAL / Digital Sundial Learning

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea / scalable experiment

What It Is: Learn the mapping needed for a digital sundial, starting with only 3 bits and seven hour states, then increasing resolution toward finer time intervals and more realistic geometry.

Tools and Concepts Used: Python; binary representation, function approximation, geometry, optimization, model complexity.

MorphoGender

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea / collaboration-ready small experiment

What It Is: Test how well models can predict grammatical gender from the written forms of Russian and French nouns, asking how much supposedly memorized gender is encoded in morphology and etymological patterns.

Tools and Concepts Used: Python; classification, feature engineering, morphology, interpretable ML, NLP.

Ambiguity Corpus

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea / larger research seed

What It Is: Build a corpus of naturally occurring and constructed ambiguities using clarification phrases, syntactic ambiguity, puns, and minimal examples. It could support ambiguity classification and tests of whether LLMs can identify and justify multiple readings.

Tools and Concepts Used: Python/regex/LLMs; corpus construction, parsing, ambiguity taxonomy, NLP evaluation.

Twain/Feynman Model Collapse

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea / staged experiment

What It Is: Start with playful Twain↔Feynman style transformation, then study what happens when models are repeatedly trained on generated output.

Tools and Concepts Used: Python/LLMs; style transfer, synthetic data, language modeling, evaluation, model-collapse experiments.

Mechanics: Lagrange / Hamilton / T-V Simulations

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea / computational-physics project

What It Is: Build small simulations that compare different formulations of classical mechanics and make their mathematical relationships visible computationally.

Tools and Concepts Used: Python; classical mechanics, ODEs, numerical methods, Lagrangian/Hamiltonian formulations.

Measuring Instruction Inconsistency

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea

What It Is: Develop quantitative ways to detect when instructions or requirements conflict with one another, especially in LLM-facing tasks.

Tools and Concepts Used: Python/LLMs; evaluation metrics, constraint consistency, NLP, experimental design.

RLHF Incomplete Ontology

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea

What It Is: Explore what happens when annotation or preference-learning tasks force judgments through an ontology that cannot represent all relevant cases.

Tools and Concepts Used: RLHF, annotation design, ontology modeling, classification, evaluation.

Design by Inverse Angular Light

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea / vision-stage

What It Is: Optics/geometry-oriented design idea centered on reasoning backward from angular light behavior to a desired physical configuration.

Tools and Concepts Used: Geometry, optics, inverse problems, computational modeling.

NLP-Oriented Symbolic Encodings for Visual Structure

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea / vision-stage

What It Is: Explore ways to encode visual structure symbolically so methods normally associated with language models can operate on spatial or visual relationships.

Tools and Concepts Used: Computer vision, symbolic representations, NLP, sequence modeling.

Symbolized Persistence Under Quantized Projection

Repository: https://github.com/bballdave025/dwb-ip-notes

Status: Idea / vision-stage

What It Is: Study which symbolic or structural information survives quantization or projection into lower-detail representations.

Tools and Concepts Used: Quantization, information theory, representation learning, dimensionality reduction.

Underground Systems Prediction Dataset

Repository: Not yet created

Status: Idea / multi-person collaboration candidate

What It Is: Build a worldwide dataset of metros, subways, undergrounds, and tube systems using construction, geology, engineering, expansion, accident, and remediation features, then test which properties or outcomes can be predicted from the others.

Tools and Concepts Used: Python/R; dataset design, regression/classification, feature engineering, correlation analysis, validation.

Translated-Author Attribution Experiment

Repository: Not yet created

Status: Idea / collaboration-ready NLP project

What It Is: Train models to distinguish two authors in their original language, then measure how author-identification confidence changes across translations. A stronger variant translates authors from different languages into each other's language and tests which stylistic signals survive.

Tools and Concepts Used: Python; authorship attribution, stylometry, translation, NLP, cross-lingual evaluation.